

Instructions

Answer all questions

Label your scripts with **Surname** and **student number**, failure to do so your script will not be marked

Number your pages and questions accordingly

Question 1

The variable **rdintens** is expenditures on research and development (R&D) as a percentage of sales. **Sales** are measured in millions of dollars. The variable **profmarg** is profits as a percentage of sales. Using the data in RDCHEM for 32 firms in the chemical industry, the following equation is estimated

$$\widehat{rdintens} = .472 + .321 \log(sales) + .050 \text{ profmarg}$$

$$(1.369) \quad (.216) \quad (.046)$$

$$n = 32, R^2 = .099.$$

1.1 Test the hypothesis that R&D intensity does not change with **sales** against the alternative that it does increase with **sales**. Do the test at the 5% and 10% levels [7].

1.2 Does **profmarg** have a statistically significant effect on **rdintens** at 10% level of significant? [5]

2. The equation estimated using 1990 data from RENTAL for 64 college towns. Let **rent** be the average monthly **rent** paid on rental units in a college town in South Africa. Let **pop** denote the total city population **avginc** the average city income, and **pctstu** is the student population as percentage of the total population.

$$\widehat{\log(rent)} = .043 + .066 \log(pop) + .507 \log(avginc) + .0056 \text{ pctstu}$$

$$(.844) \quad (.039) \quad (.081) \quad (.0017)$$

$$n = 64, R^2 = .458.$$

Test the null hypothesis that **pctstu** at 1% level has no ceteris paribus effect on monthly **rents**. [5]

3. Using a sample of 1801 employees the following earnings equations has been estimated:

$$\ln(\widehat{earnings}) = 7.059 + 0.147 \text{ educ} + 0.049 \text{ experience} - 0.201 \text{ female}$$

(0.135) (0.008) (0.007) (0.036)

$R^2 = 0.179$ $n = 1801$

3.1 Test the null hypothesis that there is no difference in expected earnings between women employees and men employees. Test this hypothesis against two-sided alternative, using the 5% significance level. [6]

4. Consider the following cubic or 3rd degree polynomial function and answer the questions that follow:

Dependent Variable: TC				
Included observations: 10				
Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	141.7667	6.375322	22.23678	0.0000
A	63.47766	4.778607	13.28372	0.0000
B ²	-12.96154	0.985665	-13.15005	0.0000
D ³	0.939588	0.059106	15.89677	0.0000
R-squared	0.998339			

4.1 Interpret the results of the above table [10]

5. Consider the following model and answer the questions that follow:

$$\hat{Y} = 1.30 + 4.91X_{1t} + 0.00123X_{2t} - 7.14X_{3t}$$

(2.38) (0.00022) (71.38)

5.1 Test the significance of the confidence @ 5% for X_{2t} [5]

6. Consider the following restricted and unrestricted equations.

$$\widehat{Price} = -14.47 + 0.976 \text{ assess}$$

(16.27) (0.049)

$n = 88$, $SSR = 165\,644.51$ $R^2 = 0.820$

$$\widehat{Price} = \text{assess} + \text{lotsize} + \text{sqrft} + \text{bdrms}$$

$n = 88$ $SSR = 209\,448.99$ $R^2 = 0.829$

6.1 Carry out the F test for the joint hypothesis [6]

6.2 Now, test $H_0 : \beta_2 = 0, \beta_3 = 0, \beta_4 = 0$ [6]

*****Good luck*****